

Deformable Torso Anatomy Education with Three-Dimensional Autostereoscopic Visualization and Free-Hand Interaction

Nan Zhang¹, Hongkai Wang², Tianqi Huang¹, Xinran Zhang¹ and Hongen Liao^{1*}

¹ Tsinghua University, China ² Dalian University of Technology, China

ABSTRACT

Virtual reality/augmented reality has advantages in immersive learning and conveying meta-information in anatomy education. In this study, we present an interactive naked-eye 3D torso anatomy education environment based on population anatomical information. In particular, we utilize the deformable anatomy models, constructed from big data of healthy adults, to convey the anatomy knowledge of truthfulness organ shapes and population anatomical variances. In addition, the proposed anatomy education system is conveyed by free-hand 3D autostereoscopic visualization, which supports multi-user and glass-free. A user study with fourteen students was implemented and demonstrates that the system is appropriate and useful for torso anatomy education.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interactive systems and tools; Applied computing—Education—Interactive learning environments

1 INTRODUCTION

Virtual reality/augmented reality (VR/AR) anatomy education is a novel teaching modality with unique advantages in depth perception, instant immersion, and 3D scene reconstruction. Various VR/AR-based anatomy teaching systems have been proposed [1, 2], and some further integrate the anatomy model with human body image [3] or typical clinical cases [4]. However, the current teaching systems are primarily based on individuals or simulation anatomical models, difficult to visually show students the population anatomical variances or realize the anatomy model personalized customization. In addition, the naked-eye 3D anatomy education only received a little attention while having potential benefits of multi-person classroom learning.

In this paper, we present an interactive naked-eye 3D torso anatomy education environment. Our environment, which provides deformable anatomy models of male and female torso organs constructed from 129 healthy adults, demonstrates the organ morphological differences through glass-free 3D visualization and mid-air gesture interaction. It overcomes the challenges in invariable anatomical shape and device-dependent observation and interaction in 3D anatomy education.

2 SYSTEM DESIGN AND IMPLEMENTATION

The proposed deformable anatomy education system mainly develops for anatomy education in classrooms, museums or surgery planning. As shown in Fig. 1, our torso anatomy education environment includes two groups of male and female deformable anatomy models and teaching interfaces. The environment is presented through the 3D autostereoscopic device and depth

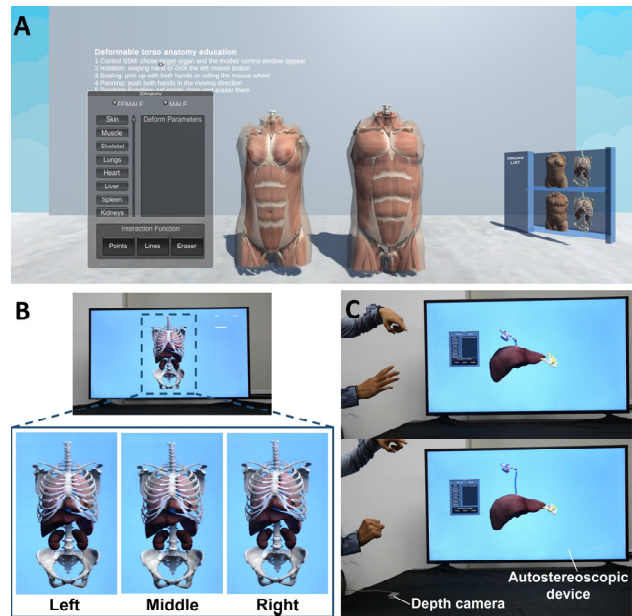


Figure 1: (A) Overview of the VR environment. (B) The prototype of the 3D autostereoscopic display and different viewpoints. (C) A user utilize mid-air gestures to add landmark and surgical planning path for the 3D liver model.

camera, providing naked-eye 3D parallax and free-hand interaction. Multiple users can observe the 3D teaching environment without wearable devices simultaneously.

2.1 Deformable Torso Anatomy Education Based on Population Variance Information

To overcome the difficulty of presenting the population anatomical variance in anatomy education, we utilize the deformable anatomy model constructed from big data in our anatomy education system. The deformable torso anatomical model includes two groups of male and female models and each has skin, muscle, trunk skeleton, lungs, heart, liver, spleen and kidneys.

We collect 129 CT images of healthy adults (diagnosed as asymptomatic) from Chinese central hospitals and construct the deformable models according to the statistical shape model (SSM) technology [5]. After manual segmentation by experienced clinicians and medical students, the organs are transformed into triangular mesh surfaces and each organ mesh registers to a standard model by sex to obtain the corresponding relationships between different samples. The corresponding relationships of the between-subjects anatomical variations are modelled as changes, recording in the deformable models. Our deformable models can be adapted from average to personalize by adjusting corresponding deformation components. Fig. 2 shows the deformable female liver model and the adaption of the male soft torso organs to an individual according to the CT image.

¹ email: nann_zhang@126.com, hanjinky@yeah.net, zhangxinran@tsinghua.edu.cn, liao@tsinghua.edu.cn

² email: wang.hongkai@dlut.edu.cn

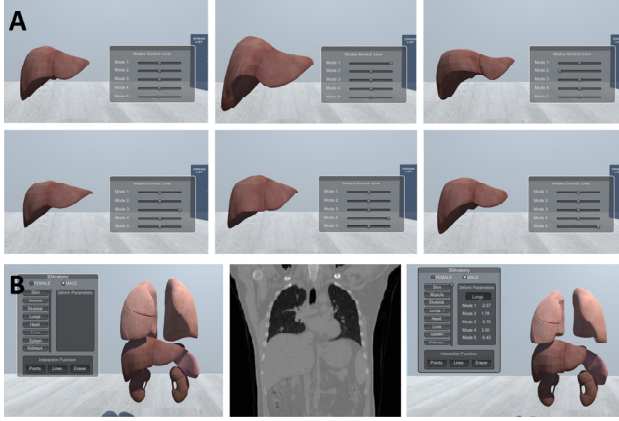


Figure 2: (A) Adjust the deformable anatomy model of female liver by deformation components. (B) The personalized anatomy adjustment, where the left, middle and right image is the average male deformable anatomy models, individual CT image, and adjusted personalized anatomy models.

2.2 3D Autostereoscopic Visualization

We utilize the 3D autostereoscopic visualization to realize the naked-eye 3D display of our system. The 3D autostereoscopic display uses the elemental image array (EIA) and lenticular lens array (LLA) to reproduce the 3D light field of a 3D scene, which provides rich parallax information and depth perception without glasses. The rendering principle can be summarized in the multi-capturing and optical reconstruction of the 3D scene. The flow work of our 3D autostereoscopic rendering includes virtual camera array setting, multi-view image array capture, pixel resampling, and EIA display. After creating the 3D anatomy education environment, ordered virtual cameras face the target scene. Each virtual camera captures its viewpoint image and is combined for one multi-viewpoint image array. The array image contains images observed from different viewpoints of the 3D environment. Subsequently, the multi-viewpoint image array is rearranged and combined into EIA. With EIA and LLA, multiple users can simultaneously observe the 3D anatomy education environment in the naked eye from different directions. Figure 1(B) shows the different viewpoints of the 3D autostereoscopic display provided by our prototype.

2.3 Free-Hand Interaction

We propose a free-hand interaction mode in mid-air to achieve system interaction. Gesture interaction in 3D space helps users control the 3D teaching environment conveniently. We develop several gestures interaction functions, including control deformable model's deformation, zoom and rotate model, move the observation field and add or delete teaching marks. To better convey the deformable anatomy knowledge, we develop two control methods based on SSM: adjusting the SSM via deformation control bar (Figure 2) and deforming the SSM through virtual hands (Position Based Dynamics algorithm). The function of dynamic zooming organs, rotating organs and controlling the movement of the observation field ensures detailed observation and learning. In addition, the teaching functions of marking landmark points or lines (Figure 1(C)) in the 3D environment help the communication between teachers and students. It also equips erasing function to make the one-click erasing of the marks. These functions allow users to learn the population anatomical information or realize personalized anatomy customization conveniently.

Table 1. Results for the user preferences questionnaires: Median (Inter-Quartile Range)

	Anatomy model	Display mode	Interaction mode
Usefulness	5.6 (5)	5.4 (5)	5.1 (5)
Easiness	5.1 (5)	4.9 (4)	4.8 (4)
Easy to use	5 (4.25)	5 (5)	4.6 (3.25)
Easy to understand	5.4 (5)	5.4 (5)	5.2 (5)
Easy to remember	5.4 (5)	5.4 (5)	5.4 (5)

3 RESULTS

The prototype (Figure 1)) is configured with a liquid crystal display panel (49.5 inches with 89.0 PPI), LLA (0.44 mm lens pitch), depth camera (Leap motion), and computer workstation (Intel Core i7-8700K CPU and 8 GB Nvidia GeForce GTX 1080).

We focus on the experience and applicability in the learning of torso anatomy based on our system. We implemented a user study with fourteen students to explore whether the proposed system has advantages of the population anatomy variance information, naked-eye 3D visualization and free-hand interaction. After introducing and freely interacting with the prototype, participants scored a questionnaire with a list of statements (6-point Likert Scale and 6 indicate full agreement). Table 1 shows the participants' reception to the proposed features of the prototype, demonstrating that all were well received and useful for torso anatomy education.

4 CONCLUSION

In this study, we present an interactive naked-eye 3D torso anatomy education environment based on Chinese population anatomical differences. There are two main contributions of the system we present: (1) we are the first to propose utilizing the deformable morphological information of the population anatomical model in VR/AR anatomy education, and achieve that through SSMs of Chinese male and female torso organs; (2) we develop the 3D autostereoscopic visualization and mid-air gesture interaction for the proposed environment, which advantages in multi-user and glass-free. The results of our user study show that our system is appropriate and useful for torso anatomy education. Our work complements the education of torso anatomy with anatomical variance knowledge and free-hand 3D autostereoscopic visualization. The proposed environment can be utilized in naked-eye 3D VR/AR display, as well as desktop PC and head mount display.

REFERENCES

- [1] R. Kurul, M. N. Ogun, A. N. Narin, S. Avci, and B. Yazgan, "An Alternative Method for Anatomy Training: Immersive Virtual Reality", *Anatomical Sciences Education*, 13(5): 648-656, 2020.
- [2] B. Preim, and P. Saalfeld, "A survey of virtual human anatomy education systems", *Computers & Graphics-Uk*, 71: 132-153, 2018.
- [3] F. Bork, L. Stratmann, S. Enssle, U. Eck, N. Navab, J. Waschke, and D. Kugelmann, "The Benefits of an Augmented Reality Magic Mirror System for Integrated Radiology Teaching in Gross Anatomy", *Anatomical Sciences Education*, 12(6): 585-598, 2019.
- [4] D. Schott, P. Saalfeld, G. Schmidt, F. Joeres, C. Boedecker, F. Huettl, H. Lang, T. Huber, B. Preim, and C. Hansen, "A VR/AR Environment for Multi-User Liver Anatomy Education", in *2021 IEEE Virtual Reality and 3D User Interfaces*, pages 296-305, 2021.
- [5] H. Wang, X. Sun, T. Wu, C. Li, Z. Chen, M. Liao, M. Li, W. Yan, H. Huang, J. Yang, Z. Tan, L. Hui, Y. Liu, H. Pan, Y. Qu, Z. Chen, L. Tan, L. Yu, H. Shi, L. Huo, Y. Zhang, X. Tang, S. Zhang, and C. Liu, "Deformable torso phantoms of Chinese adults for personalized anatomy modelling", *Journal of Anatomy*, 233(1): 121-134, 2018.